A Decay-Safe Gauged $U(1)$ Origin of the $\mathbb{Z}_{17}$ Spin(10) Axion Model

Minimal Complete-Pair Anomalons and Mass-Dependent Two-Loop Audit

Joel Ayala-Baez

Independent Researcher — jayalabaez@gmail.com

2 August 2026

Abstract

In the SO(10)×U(1)$_{\rm PQ}$ axion model of Ernst, Ringwald and Tamarit, gauging $\mathbb{Z}_{17} = U(1)_{\rm PQ}$ mod 17 requires anomaly-cancelling matter. For arbitrary vector-like $16 \oplus \overline{16}$ spectator pairs massed by the PQ singlet S, the mixed anomaly alone forces $k \equiv 5 \pmod{17}$, independent of their individual charges. Within the identical-pair ansatz all discrete anomaly classes then select the unique charge residues $\{2, 11\} \pmod{17}$. The minimal completion has five pairs, shifts the gluonic coefficient to $2N = -68$, and gives $N_{\rm DW}^{\rm cover} = 17$, $f_a = v_S/17$, unit physical domain-wall number and $E/N = 8/3$. The benchmark $v_S = M_I = 6.31 \times 10^{11}$ GeV, $y_s = \sqrt{2}$ predicts $m_a = 153.5 \pm 2.0\,\mu$eV, $\nu = 37.11 \pm 0.49$ GHz and $g_{a\gamma\gamma} = (2.335 \pm 0.125) \times 10^{-14}\,{\rm GeV}^{-1}$.

At the discrete-EFT level we distinguish a local PQ-breaking operator from a spurion set that can close into a vacuum amplitude. Although $\mathbb{Z}_{17}$ permits fermionic operators of dimensions 6, 9, 10 and 12, the renormalizable theory preserves an exact spectator vector number V. An intentionally over-complete enumeration of all monomials satisfying Lorentz parity, the Spin(10) centre and $\mathbb{Z}_{17}$ excludes a PQ-sensitive, $V = 0$ closure through $P = \sum_i (d_i - 4) = 11$. The bound is saturated at $P = 12$ by four dimension-6 portals plus one PQ-conserving dimension-8 vector-breaking spurion. An explicit 32-dimensional Clifford-algebra and Grassmann audit finds nonzero Spin(10) and Lorentz contractions, verifies Fermi statistics, and identifies a connected two-loop graph. Wilsonian NDA gives $|\Delta\bar{\theta}| \lesssim 9.04 \times 10^{-28}|C_{\rm eff}|$. The extra $\mathbb{Z}_3$ proposed in v15 is unnecessary and is removed.

We then construct a decay-safe anomaly-free origin of the discrete symmetry. Gauge $U(1)_X$, break it with a PQ-neutral scalar Φ_{17}, and add three complete pairs with charges $(1, 16)$, $(14, 3)$ and $(1, -18)$. They contribute $(+34, +272, +16592)$ to the mixed Spin(10), gravitational and cubic anomalies, exactly cancelling the light sector, while their accidental-PQ anomaly vanishes. Each pair has a renormalizable portal to ordinary 16_F matter through an already-present 10_H or S. Mixed-anomaly cancellation forces the number of single-Φ-massed pairs to be odd; one pair would require roots of $t^2 - 17t + 76$, whose discriminant is -15. Thus three is minimal in this complete-pair-only ansatz (with no Spin(10)-singlet anomalons), and exact enumeration finds the displayed triple uniquely in the stated existing-field portal basis.

The decay portals change, rather than invalidate, the quality audit. A heavy-inclusive necessary-condition enumeration excludes a vacuum closure through $P = 7$ and is saturated at $P = 8$ by four dimension-five portals plus one dimension-eight spectator spurion. An explicit nonzero connected two-loop graph produces $\Phi^4 (S^\dagger)^{18} (10_H^\dagger)^2$. Its repeated-pole integral retains all three masses and gives $2|A_8|/\chi = 6.04 \times 10^{-47}|C_8|$ at $v_\Phi = 10^{17}$ GeV. The direct dimension-21 scalar operator remains the largest explicitly computed unit-coefficient term, $2|A|/\chi = 4.60 \times 10^{-43}$. A normalized decay portal as small as 3.1×10^{-20} gives a lifetime below one second; every 16 component has a nonzero 10_H channel in the explicit Clifford representation. The construction is an anomaly-free field theory to the reduced Planck cutoff, not a string completion; flavour/Clebsch fits, hierarchy stabilization and the string network remain open. Three fail-closed engines perform 65, 59 and 42 checks; the release adds independent charge, decay and loop-quadrature tests.

1 Model and conventions

Fields and PQ charges: 16_F (+1, three generations), 10_H (-2), $\overline{126}_H$ (-2), 210_H (0), S (+4) [1]; the axion is dominantly $\arg S$, phase-locked by $\overline{126}_H^2 10_H^2 S^2$. We gauge $\mathbb{Z}_{17} := \mathrm{U}(1)_{\mathrm{PQ}} \bmod 17$. Spectators: k identical $16_s \oplus \overline{16}_s$ pairs with $\mathbb{Z}_{17}$ charges (a, b), masses from $S\,16_s\overline{16}_s$, i.e. thresholds $M_s = y_s v_S/\sqrt{2}$. Discrete-gauge protection of axions has a long history [6, 7, 8, 9]; the specific result below concerns this particular $\mathbb{Z}_{17}$ extension of the ERT model.

2 The theorem

Theorem 1 (minimality and residue uniqueness, identical-pair ansatz)**.** *The $\mathbb{Z}_{17}$ anomaly classes (mixed SO(10)^2, gravitational, cubic [24, 2]) all vanish modulo 17 if and only if $k \equiv 5 \pmod{17}$ and $\{a, b\} \equiv \{2, 11\} \pmod{17}$. The minimal solution therefore contains five pairs, and the charge* residues *are unique up to interchange;* $(2, -6)$ *is one integer lift.*

Proof. The mass operator forces $a + b \equiv -4 \equiv 13$. The mixed class is $6 + 2k(a+b) \equiv 6 - 8k$; with $8^{-1} \equiv 15$, vanishing requires $k \equiv 90 \equiv 5$, i.e. $k = 5 + 17m$. The gravitational class $48 + 16k(a+b) \equiv 14 + 4k$ vanishes automatically on the same set. The cubic class uses $a^3 + b^3 = (a+b)^3 - 3ab(a+b) \equiv 4 - 5ab$, giving $48 + 80(a^3+b^3) \equiv 11 - 9ab \equiv 0$, i.e. $ab \equiv 5$. The residue pair is then the root set of $x^2 - 13x + 5 \equiv 0 \pmod{17}$, whose only roots are 2 and 11; an exhaustive residue scan confirms uniqueness. □

The five-pair lower bound is stronger than the theorem's identical-pair wording. For arbitrary residues $(a_i, 13 - a_i)$, the mixed class is

$$6 + 2\sum_{i=1}^{k}\big[a_i + (13 - a_i)\big] \equiv 6 + 9k \pmod{17}, \tag{1}$$

so every charge assignment requires $k \equiv 5 \pmod{17}$. Direct enumeration gives no solution for $k \leq 4$ and 83 232 ordered cubic solutions among the 17^5 tuples at $k = 5$. Thus pair-count minimality is charge-independent, whereas the residue uniqueness $\{2, 11\}$ is specifically an identical-pair result. The general cubic class used here is $48 + 16\sum_i[a_i^3 + (13 - a_i)^3]$; the alternate engine incorrectly used 80 per pair (valid only after five identical pairs), although its solution count happened to remain unchanged.

The five pairs shift the gluonic coefficient to $2N = 2[6 + 10(a+b)] = -68$, so

$$N_{\mathrm{DW}}^{\mathrm{cover}} = \frac{|2N|}{X_S} = \frac{68}{4} = 17, \qquad f_a \longrightarrow \frac{v_S}{17}, \tag{2}$$

in the discrete EFT, or equivalently the $v_\Phi \gg v_S$ limit of Section 5; the exact continuous-theory projection is given in Eq. (18). The ratio $E/N = 8/3$ is preserved (complete SO(10) multiplets). All seventeen covering-space minima form one gauge orbit ($4 \times 13 \equiv 1 \bmod 17$): the physical domain-wall number is one, and the one-wall object is the string sector $(\ell, n) = (13, -3)$ with S-winding $2\pi/17$ (we reserve k for the pair multiplicity).

3 Benchmark and inputs

One-loop running of $\alpha_1^{-1}(M_Z) = 59.02$ (GUT-normalised), $\alpha_2^{-1}(M_Z) = 29.57$, $\alpha_s(M_Z) = 0.1179$ with 2HDM coefficients $(21/5, -3, -7)$ below M_I, Pati–Salam coefficients $(-7/3, 2, 26/3)$ above it, and matching $\alpha_{2R}^{-1} = \frac{5}{3}\alpha_1^{-1} - \frac{2}{3}\alpha_4^{-1}$ gives $M_I = 6.31 \times 10^{11}$ GeV, $M_{\mathrm{GUT}} = 9.92 \times 10^{15}$ GeV, $\alpha_{\mathrm{GUT}}^{-1} = 37.31$. *Gauge unification does not fix the singlet scale [1]; we separately impose the* *benchmark identifications $v_S = M_I$ and $y_s = \sqrt{2}$* (so $M_s = M_I$). Then, with $\chi^{1/4} = 75.5(5)$ MeV and the QCD photon coefficient 1.92(4) from Grilli di Cortona et al. [20],

$$m_a = \frac{\chi^{1/2}}{f_a} = 153.5 \pm 2.0\,\mu\text{eV}, \qquad \nu = \frac{m_a}{h} = (37.11 \pm 0.49)\ \text{GHz}, \tag{3}$$

$$g_{a\gamma\gamma} = \frac{\alpha}{2\pi f_a}\left|\frac{E}{N} - 1.92\right| = (2.335 \pm 0.125) \times 10^{-14}\,\text{GeV}^{-1}, \tag{4}$$

where the mass and frequency uncertainties propagate the susceptibility error and the coupling uncertainty propagates the 1.92(4) coefficient. The compatible determination $\chi^{1/4} = 75.6(1.8)(0.9)$ MeV [19] would widen the mass uncertainty to about $8\,\mu$eV. The spectator shift is $\Delta\alpha^{-1}_{\text{GUT}} = -\frac{40/3}{2\pi}\ln(M_{\text{GUT}}/M_s) = -20.5$ at the benchmark y_s; other y_s values shift it (e.g. -26.1 at $y_s = 0.1$), so y_s is an explicit input of the proton estimate.

4 Axion quality: local operators and vacuum closure

At the reduced Planck scale we assume every local operator allowed by the gauge group is present, with a dimensionless Wilson coefficient of order unity, and that no intermediate threshold carries additional explicit PQ breaking. This assumption is essential: heavy fields can modify naive Planck power counting [11].

Scalar sector and the local-operator trap

The 10_H and $\overline{126}_H$ tensors have an odd number of SO(10) vector indices, whereas 210_H has an even number. Deltas and the rank-ten epsilon tensor contract an even total number of vector indices, so an invariant contains an even number of odd-index tensors. Including S (charge $+4$), every scalar invariant therefore has $Q_{\text{PQ}} = 0 \pmod 4$. Combining this with $\mathbb{Z}_{17}$ gives $|Q_{\text{PQ}}| \geq \text{lcm}(4, 17) = 68$, saturated by $S^{17}/M_{\text{Pl}}^{13}$. At the benchmark, the one-sided coefficient is $(v_S/\sqrt{2})^{17}/(M_{\text{Pl}}^{13}\chi) = 3.24 \times 10^{-37}$. With $V = Ae^{ia/f_a} + \text{h.c.}$ the worst-phase shift is

$$\frac{|\Delta\bar{\theta}|_{S^{17}}}{|c_{17}|} = \frac{2(v_S/\sqrt{2})^{17}}{M_{\text{Pl}}^{13}\chi} = 6.47 \times 10^{-37}, \qquad M_{\text{Pl}} = 2.435 \times 10^{18}\,\text{GeV}. \tag{5}$$

Fermions invalidate any proof that stops there; multi-fermion operators are known to matter in axion-quality analyses [10]. Write $F = 16_F$, $s = 16_s$ and $b = \overline{16}_s$, with $Q_{\text{PQ}} = (1, 2, -6)$. SO(10)$\times\mathbb{Z}_{17}$ permits, among others,

$$\begin{aligned}
\mathcal{O}_6^- &= \frac{(S^\dagger)^3(Fb)_1}{M_{\text{Pl}}^2}, && (d, Q, V) = (6, -17, -1),\\
\mathcal{O}_8^0 &= \frac{(S^\dagger)^2[(ss)_{10}]^2}{M_{\text{Pl}}^4}, && (d, Q, V) = (8, 0, +4),\\
\mathcal{O}_9 &= \frac{(S^\dagger)^5(bb)_{10}10_H}{M_{\text{Pl}}^5}, && (d, Q, V) = (9, -34, -2),\\
\mathcal{O}_{10} &= \frac{10_H^\dagger[(bb)_{10}]^3}{M_{\text{Pl}}^6}, && (d, Q, V) = (10, -34, -6),\\
\mathcal{O}_{12} &= \frac{(S^\dagger)^6(Fs)_{10}(ss)_{10}}{M_{\text{Pl}}^8}, && (d, Q, V) = (12, -17, +3),
\end{aligned} \tag{6}$$

where $V(s) = +1$, $V(b) = -1$ and all ERT fields have $V = 0$. The representation products used in these explicit contractions are standard, in particular $16 \otimes 16 = 10_s \oplus 120_a \oplus 126_s$ [12]. The dimension-10 and -12 terms are regression cases missed by a bilinear-only enumeration.

Nor does $\mathcal{O}_9$ by itself generate the previously claimed term linear in its Majorana insertion. For one Dirac pair,

$$\mathcal{M} = \begin{pmatrix} 0 & M_s \\ M_s & \delta e^{i\phi} \end{pmatrix}, \quad \mathrm{tr}(\mathcal{M}^\dagger \mathcal{M}) = 2M_s^2 + |\delta|^2, \quad \det(\mathcal{M}^\dagger \mathcal{M}) = M_s^4. \tag{7}$$

Both singular values, hence the one-loop Coleman–Weinberg potential, are independent of ϕ. A local PQ-breaking operator is therefore not automatically a one-insertion axion potential.

Finite multi-spurion proof

The renormalizable Lagrangian preserves the spectator vector number $U(1)_V$: the mass Ssb is neutral and there is no dimension-four or lower $Q_{\mathrm{PQ}} = 0$, $V \neq 0$ interaction. Consequently a vacuum graph must have total $V = 0$. Each Planck insertion has $Q_{\mathrm{PQ}} = 17m$; after its fermions are closed with PQ-conserving vertices, scalar index parity also requires total $Q_{\mathrm{PQ}} = 0 \pmod 4$. A PQ-sensitive target thus has

$$\sum_i V_i = 0, \qquad \sum_i m_i \neq 0, \qquad \sum_i m_i = 0 \pmod 4. \tag{8}$$

The supplied engine enumerates a deliberate *superset* of local monomials: F, s, b and all conjugates; $S, S^\dagger$; charge-∓ 2 tensor scalars; even Weyl count; trivial Spin(10)-centre charge; and $Q_{\mathrm{PQ}} = 0 \pmod{17}$. It allows mixed-chirality contractions that may not exist, so failure to find a closure is a conservative exclusion. Neutral 210_H, derivatives and gauge strengths can be omitted because they only raise dimension. For

$$P := \sum_i (d_i - 4) \leq 11, \tag{9}$$

every insertion has $d_i \leq 15$; enumeration through dimension 15 is therefore finite and complete for this exclusion. No state satisfying Eq. (8) occurs through $P = 11$. At $P = 12$ the bound is saturated by the actual Spin(10) set

$$4\mathcal{O}_6^- + \mathcal{O}_8^0: \qquad (Q_{\mathrm{PQ}}, V, P) = (-68, 0, 12). \tag{10}$$

The necessary-condition search was independently reimplemented directly from weak compositions of field multiplicities; it reproduces all twenty low-cost $(P, Q_{\mathrm{PQ}}/17, V)$ frontier classes and the same absence result through $P = 11$ without importing the main enumeration module.

Explicit Spin(10), Lorentz and graph audit

Because spinors are representations of the double cover, this audit uses Spin(10) explicitly ("SO(10)" elsewhere is the standard shorthand). Let Γ^a be a 32×32 Euclidean Clifford representation and $C = \Gamma^2\Gamma^4\Gamma^6\Gamma^8\Gamma^{10}$. On either chiral 16-dimensional subspace, the ten matrices

$$T_\pm^a = (C\Gamma^a)|_{16_\pm}, \qquad (T_\pm^a)^T = T_\pm^a, \qquad \sum_{ij} (T_+^a)^*_{ij} (T_+^b)_{ij} = 16\delta^{ab} \tag{11}$$

are symmetric, so the same-chirality Weyl bilinear $\epsilon_{\alpha\beta} X_i^\alpha (T_\pm^a)_{ij} Y_j^\beta$ has the required Fermi statistics for the 10_s channel. Sparse Grassmann evaluation gives nonzero polynomials for every operator in Eq. (6); in particular $\mathcal{O}_8^0$ has 240 nonzero canonical monomials even when its four s fields share a species.

For the central four-spinor tensor $K_{ijkl} = \sum_a (T_+^a)_{ij} (T_+^a)_{kl}$, the explicit representation gives 640 nonzero entries and

$$\sum_{ijkl} |K_{ijkl}|^2 = 2560, \qquad K^*_{ijkl} (T_+^1)_{ij} (T_+^1)_{kl} = 256, \qquad L_{\mathrm{Lorentz}} = 4. \tag{12}$$

Thus the certificate is not merely a centre-charge candidate: its actual group and Lorentz contractions are nonzero. Four s–b same-chirality propagators (the four s legs may be assigned four of the five spectator copies) and two ordinary-family 10_H-channel chirality flips close the twelve fermions. With five compact vertices and six internal fermion lines, $L = 6 - 5 + 1 = 2$. The resulting scalar operator is proportional to

$$(S^\dagger)^{18}(10_H^\dagger \cdot 10_H^\dagger), \qquad Q_{\rm PQ} = -68. \tag{13}$$

A special Higgs alignment could make the dot product vanish, which would only strengthen the quality bound by moving the first nonzero vacuum term to higher order.

Treating all masses and scalar insertions as no larger than the matching scale M_s and dropping loop suppressions gives the Wilsonian-NDA upper bound

$$\frac{|\Delta\bar\theta|}{|C_{\rm eff}|} \lesssim \frac{2M_s^4}{\chi}\left(\frac{M_s}{M_{\rm Pl}}\right)^{12} = 9.04 \times 10^{-28}. \tag{14}$$

Here $C_{\rm eff}$ absorbs convention-dependent tensor normalisations, Yukawas and flavour factors. For the explicit graph, retaining the two loop factors and using $|v_{10}| \le 246\,\text{GeV}$ yields the diagnostic

$$\frac{|\Delta\bar\theta|_{P=12}}{|C_{\rm eff}|} \lesssim 9.04 \times 10^{-28}(16\pi^2)^{-2}\left(\frac{246\,\text{GeV}}{M_s}\right)^2 = 5.50 \times 10^{-51}, \tag{15}$$

before any additional flavour suppression. Equation (14), which deliberately omits these small factors, remains the advertised conservative result.

Proposition 1 (quality under the stated EFT assumptions)**.** *At the benchmark $M_s = v_S = 6.31 \times 10^{11}\,\text{GeV}$, the minimal SO(10)$\times\mathbb{Z}_{17}$ completion satisfies the neutron-EDM requirement $|\Delta\bar\theta| < 10^{-10}$ provided $|C_{\rm eff}| < 1.1 \times 10^{17}$. In particular, order-one Wilson coefficients are safe; the leading conservative bound is Eq. (14), not the scalar-only value in Eq. (5).*

5 An anomaly-free continuous origin of $\mathbb{Z}_{17}$

A discrete gauge symmetry can arise as the remnant of a broken continuous gauge group [3, 4, 5]. Introduce a complex Spin(10) singlet Φ with $(X, \text{PQ}) = (17, 0)$ and gauge a primitive $U(1)_X$. The light-field X charges are the integer PQ lifts used above: $X(F, s, b, S, 10_H, \overline{126}_H, 210_H) = (1, 2, -6, 4, -2, -2, 0)$. Replace the relic-unsafe v19 anomalons by the left-Weyl pairs

$$(P, \overline{P}) \sim (16_1, \overline{16}_{16}), \qquad (Q, \overline{Q}) \sim (16_{14}, \overline{16}_3), \qquad (R, \overline{R}) \sim (16_1, \overline{16}_{-18}). \tag{16}$$

Their accidental-PQ charges are respectively $(1, -1)$, $(-3, 3)$ and $(1, -1)$. Renormalizable masses and decay portals are

$$\Phi^\dagger(y_P P\overline{P} + y_Q Q\overline{Q}) + y_R \Phi R\overline{R} + \lambda_P PF10_H + \lambda_R RF10_H + \lambda_Q \overline{Q}FS^\dagger + \text{h.c.} \tag{17}$$

Every X and PQ charge sum in Eq. (17) vanishes exactly. The two added 16_1 fields share the ordinary-family gauge charges. A generic rank-two Φ mass matrix pairs two linear combinations of the five 16_1 fields with $\overline{16}_{16}$ and $\overline{16}_{-18}$, leaving exactly three chiral light combinations. Thus the labels F, P, R denote a convenient UV basis, not a claim that the heavy–light mass matrix is diagonal.

With $T(16) = 2$ and $\dim 16 = 16$, the continuous anomaly audit is

sector	A_{10^2X}	A_{grav^2X}	A_{X^3}
light fields	-34	-272	-16592
three pairs in Eq. (16)	$+34$	$+272$	$+16592$
total	0	0	0

The cubic entries before the common factor 16 are $4097 + 2771 - 5831 = 1037$. The heavy accidental-PQ mixed anomaly is also zero, so the axion's QCD anomaly and low-energy couplings are unchanged.

There is a short minimality proof for an anomalon sector built only from complete pairs, with no Spin(10)-singlet anomalons. If every $16 \oplus \overline{16}$ pair receives a renormalizable mass from one Φ or $\Phi^\dagger$, its two X charges sum to ± 17. Cancellation of the mixed anomaly requires $n_+ - n_- = 1$, hence an odd number of pairs. For one pair, $x + y = 17$ and cubic cancellation require $x^3 + y^3 = 1037$, or $xy = 76$. The roots would satisfy $t^2 - 17t + 76 = 0$, whose discriminant is -15. One pair is impossible, whereas Eq. (16) supplies three; three is therefore minimal under these assumptions. Exact enumeration of portals constructed from the already-present 10_H, S and 210_H finds the displayed triple as the unique three-pair solution. This is not a classification involving arbitrary Spin(10) representations, singlet anomalons or multi-scalar mass operators.

An over-catalogue of every dimension-four-or-lower monomial satisfying Lorentz parity, the Spin(10) centre and exact $U(1)_X$ contains no term that breaks accidental PQ or the light-spectator vector number. The heavy pair numbers are explicitly broken by Eq. (17), as required for decay. Hence PQ remains accidental but no exact anomalon number makes a heavy state stable. PQ is independent of the gauged generator because Φ has $X = 17$ but PQ zero. Writing the canonically normalized phases as (a_Φ, a_S), define

$$D = \sqrt{(17v_\Phi)^2 + (4v_S)^2}, \qquad \widehat{a} = \frac{-4v_S a_\Phi + 17 v_\Phi a_S}{D}, \qquad f_a = \frac{v_S v_\Phi}{D}. \tag{18}$$

The first combination is orthogonal to the gauge direction. For $v_\Phi = 10^{17}$ GeV the correction to $v_S/17$ is -1.10×10^{-12}.

The large charges make the Abelian running a real constraint. Including all fields above v_Φ gives

$$b_X = \frac{2}{3}\sum_{\text{Weyl}} d(R)X^2 + \frac{1}{3}\sum_{\text{scalar}} d(R)X^2 = \frac{2}{3}(15840) + \frac{1}{3}(849) = 10843. \tag{19}$$

Requiring the one-loop Landau pole to exceed $M_{\text{Pl}} = 2.435 \times 10^{18}$ GeV at $v_\Phi = 10^{17}$ GeV requires $g_X(v_\Phi) < 0.0478$; $g_X = 0.04$ gives $\Lambda_{\text{LP}} = 9.47 \times 10^{18}$ GeV. Complete Spin(10) multiplets do not change one-loop relative unification below their common threshold. Using $T(10, 126, 210) = (1, 35, 56)$ [13] and conservatively treating even the real 210_H as complex gives $b_{10} \le 80/3$ above v_Φ. An older diagnostic reset $\alpha_{10}(v_\Phi) = 1/40$ would remain perturbative, but that reset is inconsistent with the model's own spectator-corrected $\alpha_{\text{GUT}}^{-1} \simeq 16.81$. Continuous one-loop running from that value is *not* weakly coupled to M_{Pl} under these betas (the conservative envelope hits a Landau pole below M_{Pl}). A full two-loop threshold analysis is therefore required before claiming Planck-safe Spin(10) perturbativity.

The decay statement is quantitative. For one open normalized two-body channel with final mass m_ϕ,

$$\Gamma = \frac{|\lambda|^2 M}{32\pi}\left(1 - \frac{m_\phi^2}{M^2}\right)^2 \le \frac{|\lambda|^2 M}{32\pi}, \qquad M = \frac{v_\Phi}{\sqrt{2}}. \tag{20}$$

The massless formula is therefore an upper benchmark on the width. At the benchmark, $\lambda = 10^{-8}$ and $m_\phi \to 0$ gives $\tau = 9.36 \times 10^{-24}$ s, while $\tau < 1$ s in that massless limit requires only $|\lambda| > 3.06 \times 10^{-20}$. In the explicit Clifford basis, $\sum_{a,j} (C\Gamma^a)_{ij} (C\Gamma^a)^*_{kj} = 10\delta_{ik}$, proving that every component of P and R has a nonzero 10_H channel. The $\overline{Q} F S^\dagger$ singlet contraction is component-diagonal. This removes the exact-stable-relic obstruction; a broken-phase Clebsch and flavour fit is still required for precision widths.

Finally, the portal $|\Phi|^2|S|^2$ must be small enough to preserve $v_\Phi \gg v_S$. These are explicit perturbativity and hierarchy assumptions, so this is a field-theory completion to the Planck cutoff rather than an ultimate quantum-gravity construction.

6 Heavy-threshold closure and full loop kernels

Heavy fields can invalidate naive axion-quality power counting [11], so the operator search was repeated with all fields in Eq. (16) and their conjugates. Because the decay vertices break the heavy pair numbers, only the exact light-spectator number V_s is imposed. The resulting minimum-dimension frontier has no PQ-sensitive, $V_s = 0$ closure through $P = 7$. At $P = 8$ the necessary-condition bound is saturated by actual Spin(10) invariants:

$$\begin{aligned}
\widetilde{\mathcal{O}}_5 &= \frac{(S^\dagger)^2(Qb)_1}{M_{\rm Pl}}, && (Q_{\rm PQ}, V_s, P) = (-17, -1, 1), \\
\widetilde{\mathcal{O}}_8 &= \frac{(S^\dagger)^2[(ss)_{10}]^2}{M_{\rm Pl}^4}, && (Q_{\rm PQ}, V_s, P) = (0, +4, 4).
\end{aligned} \tag{21}$$

The combination $4\widetilde{\mathcal{O}}_5 + \widetilde{\mathcal{O}}_8$ closes with four chirality-flipping s–b propagators generated by Ssb, four Q–$\overline{Q}$ propagators generated by $\Phi^\dagger Q\overline{Q}$, four $\overline{Q}FS^\dagger$ decay vertices and two conjugate ordinary $FF10_H$ Yukawas. The left–left propagators carry the conjugate mass spurions, $S^\dagger$ and Φ, respectively. Counting the five Planck vertices, four decay vertices and two ordinary Yukawas gives a connected graph with 12 internal fermion lines, 11 vertices and two loops. It produces

$$\Phi^4(S^\dagger)^{18}(10_H^\dagger)^2, \qquad (X, Q_{\rm PQ}, P) = (0, -68, 8). \tag{22}$$

The explicit Clifford representation gives $(T^a)^*_{ij}(T^b)_{ij} = 16\delta^{ab}$ and the inherited four-bilinear two-component contraction, including both kinetic numerators, equals -2 per factorized loop (hence 4 for their product), so this saturation graph is nonzero.

The original v17 two-loop graph also survives, but its four $\mathcal{O}_6^-$ portals each carry $X = -17$. Exact gauge invariance replaces each by $\Phi\mathcal{O}_6^-/M_{\rm Pl}$; the graph is therefore $P = 16$. The full zero-momentum integral can be written without dimensional estimates. For degenerate spectator mass M_s and a family chirality mass m,

$$\begin{aligned}
I_3(M_s, m) &= \int \frac{d^4p_E}{(2\pi)^4} \frac{1}{(p^2 + M_s^2)^2(p^2 + m^2)} \\
&= \frac{1}{16\pi^2 M_s^2} \frac{1 - r + r\ln r}{(1-r)^2}, \qquad r = \frac{m^2}{M_s^2},
\end{aligned} \tag{23}$$

$$\begin{aligned}
J(M_s, m) &= M_s^2 m\, I_3(M_s, m), \\
|A_{2\ell}^{\rm EFT}| &= |C_{2\ell}| \frac{s_0^{14}}{M_{\rm Pl}^{12}} J(M_s, m_1) J(M_s, m_2), \qquad s_0 = \frac{v_S}{\sqrt{2}}.
\end{aligned} \tag{24}$$

For unequal masses the code evaluates the symmetric divided difference

$$I_3 = \frac{1}{16\pi^2} \sum_x \frac{x\ln x}{\prod_{y\neq x}(x-y)}. \tag{25}$$

Equation (24) is finite and factorizes because the zero-momentum four-fermion vertex leaves two independent loop momenta. The normalized coefficient $C_{2\ell}$ contains the unknown Planck Wilson tensors, flavour contractions, the nonzero group/Lorentz normalization and RG matching; the EFT cannot predict it.

For the $P = 8$ graph, each of the two factorized loops has the independently evaluated kernel

$$K(M_H, M_s, m_f) = M_H^2 M_s^2 \int \frac{d^4p_E}{(2\pi)^4} \frac{p^2}{(p^2 + M_H^2)^2(p^2 + M_s^2)^2(p^2 + m_f^2)^2}. \tag{26}$$

High-precision repeated-pole partial fractions retain all three masses and are independently checked by log-coordinate quadrature. The one-sided amplitude is

$$|A_8| = |C_8| \frac{s_0^{14} h^2}{M_{\rm Pl}^8} K(M_H, M_s, m_f)^2. \tag{27}$$

Here C_8 contains the four decay Yukawas, two family Yukawas and the normalized Wilson, group, flavour and RG contraction. With $M_s = v_S$, $M_H = v_\Phi/\sqrt{2}$, $m_f = h = 246\,\mathrm{GeV}$ and $v_\Phi = 10^{17}\,\mathrm{GeV}$, $16\pi^2 M_H^2 K = 0.9999999965$. The results per unit normalized coefficient are

term	$\lvert A\rvert/\chi$	worst phase $2\lvert A\rvert/\chi$
v17 two-loop EFT graph, before Φ dressings	2.15×10^{-53}	4.30×10^{-53}
same graph in $U(1)_X$ ($P = 16$)	1.53×10^{-59}	3.06×10^{-59}
decay-threshold graph ($P = 8$)	3.02×10^{-47}	6.04×10^{-47}
direct $\Phi^4(S^\dagger)^{17}/M_{\rm Pl}^{17}$	2.30×10^{-43}	4.60×10^{-43}

The direct dimension-21 scalar operator dominates these explicitly computed terms and still lies 2.17×10^{32} below the 10^{-10} criterion for a unit coefficient. This numerical margin is conditional on order-one normalized Wilson contractions; it is not a measurement or a derivation of quantum-gravity coefficients.

An extra vector-like spectator $\mathbb{Z}_3$ is neither needed nor used. Indeed $\mathcal{O}_{10}$ and $\mathcal{O}_{12}$ survive it, falsifying the earlier claim that all fermionic breakers move to dimension 19. Such an exact triality would also make the lightest triality-charged state stable unless further structure were added. The rank-six rephasing diagnostic in Appendix A remains correct but is not the quality proof. The continuous origin is supplied in Section 5; its cosmic-string spectrum remains open.

Adversarial merge audit of the alternate anomalon lift

The alternate charges $16_1 \oplus \overline{16}_{16}$ and singlets $(33, -16, 31, -48)$ do cancel the three continuous anomaly sums exactly. They do not, however, establish the advertised quality theorem. A physical axion first requires a global accidental PQ independent of the gauged generator; one possible repair assigns $Q_{\rm PQ}(16_1, \overline{16}_{16}, \Phi) = (1, -1, 0)$. With that assignment the following actual 10-channel invariants are allowed:

$$
\begin{aligned}
calO_6^A &= \frac{(S^\dagger)^2(\overline{16}_{16}\,\overline{16}_{s,-6})_{10}10_H}{M_{\rm Pl}^2}, && (Q_{\rm PQ}, V_s, P) = (-17, -1, 2),\\
calO_8^s &= \frac{(S^\dagger)^2[(16_{s,2}16_{s,2})_{10}]^2}{M_{\rm Pl}^4}, && (Q_{\rm PQ}, V_s, P) = (0, +4, 4).
\end{aligned}
\tag{28}
$$

Hence $4\mathcal{O}_6^A + \mathcal{O}_8^s$ is a $(Q_{\rm PQ}, V_s, P) = (-68, 0, 12)$ closure. The proposed $16_1 16_F 10_H$ vertex breaks anomalon number, and four anomalon-mass propagators, four such vertices and two ordinary $16_F 16_F 10_H$ vertices provide a connected two-loop topology for the remaining fermions. An end-to-end Grassmann contraction of that alternate topology is still required. The scalar-only dimension-21 scan therefore misses the relevant fermionic threshold; this does not prove that the alternate model is unsafe, but it does mean its published quality bound and loop amplitude are incomplete.

The same audit finds only the two Φ mass Yukawas for the four heavy singlets, not a renormalizable portal to light matter. It also gives $b_X = 8263$ and $g_X(M_{\rm GUT}) < 0.0417$ for a Landau pole above $M_{\rm Pl}$. Finally, applying the exact factorized kernel to the *old* graph at the alternate inputs gives $\lvert A\rvert/\chi = 1.85 \times 10^{-64}$ after four dressings, rather than 2.37×10^{-62}; the ratio is the earlier 128-fold vev-normalization envelope. These are the reasons the alternate anomalon sector is not merged, while its charge-general minimality proof is retained.

7 Status of cosmology, experiment, and proton decay

Cosmology (conditional). The v20 anomalons carry no exact heavy number and decay through Eq. (17); unlike v19, avoiding a stable heavy relic does not require a low-reheat postulate. This statement does not replace a Boltzmann calculation if the $U(1)_X$ sector is thermally restored. Pre-inflationary: $\theta_i = 2.91$ on the anharmonic tail [15], $H_I \lesssim 9 \times 10^5\,\mathrm{GeV}$ (UV-conditional [16]), corresponding to $r < 1.34 \times 10^{-17}$ for $A_s = 2.1 \times 10^{-9}$. A primordial B-mode detection above that ceiling rejects this pre-inflationary all-dark-matter branch, not the post-inflationary branch. The latter is conditional on the $(\ell, n) = (13, -3)$ string's existence and production [14]. If a standard network applied, current simulations disagree on the full-DM mass: Benabou et al. find 45–65 μeV before wall collapse, with wall effects potentially extending to $\sim 300\,\mu$eV [17]; Saikawa et al. report 95–450 μeV [18]. The benchmark lies inside the latter and inside the extended range of the former; the nonstandard gauged network makes all such intervals indicative.

Experiment. No dark-matter haloscope has scanned 153.5 μeV: the recent DALI results sit at 28.5 μeV [22] (their reported sensitivity is disputed [23]), and no existing experiment reaches the predicted coupling there; CAST constrains the mass at $5.8 \times 10^{-11}\,\mathrm{GeV}^{-1}$ [25], a factor 2 479 above. Direct tests belong to MADMAX [26] and plasma-haloscope [27] programmes; the ORGAN design range also overlaps the target [28]. Their design ranges are not completed scans or guaranteed DFSZ sensitivity. The benchmark location is 36.62–37.60 GHz, while the Galactic signal width is only about 34 kHz for $Q_{\mathrm{halo}} = 1.1 \times 10^6$. A null scan of the full location band below $g_{a\gamma\gamma} = 2.335 \times 10^{-14}\,\mathrm{GeV}^{-1}$ would reject the all-dark-matter benchmark combination ($v_S{=}M_I$, $y_s{=}\sqrt{2}$), not the anomaly theorem itself.

Proton decay (prior-region statements). Under the diagnostic priors of Table 1, the median lifetime is 1.1×10^{35} yr; 35 per cent of sampled volume lies below the Super-K cutoff 2.4×10^{34} yr [29] and a further 14 per cent below the Hyper-K reach 1×10^{35} yr [30]; a Hyper-K null would exclude only that additional subregion.

Table 1: Diagnostic priors of the proton Monte Carlo (seed 20260801, 10^5 trials; spectator shift applied per trial with $y_s = \sqrt{2}$).

$\alpha_s(M_Z)$	$\mathcal{N}(0.1179,\ 0.0009)$
$\alpha_1^{-1},\ \alpha_2^{-1}$	$\mathcal{N}(59.02, 0.02)$, $\mathcal{N}(29.57, 0.02)$
threshold	$\Delta_4 = \delta \sim \mathcal{U}(-1, 1)$, $\Delta_{2R} = -\frac{2}{3}\delta$, $\Delta_{2L} = 0$
M_X/M_{GUT}	LogU(0.5, 2)
A_R	$\mathcal{U}(2.0,\ 3.2)$
W_0 [GeV^2]	$\mathcal{N}(0.1095,\ 0.011)$

8 Novelty and conclusions

The defensible contribution now has five layers: (i) the charge-independent five-pair lower bound, plus residue uniqueness in the identical-pair ansatz, for this particular $\mathbb{Z}_{17}$ extension; (ii) the discrete-EFT $P = 12$ multi-spurion lower bound and its explicit Spin(10) graph; (iii) the anomaly-free $U(1)_X \to \mathbb{Z}_{17}$ field-theory embedding of Section 5; (iv) the one-pair no-go and minimal three-pair decay-safe completion; and (v) the heavy-inclusive $P = 8$ closure audit plus its repeated-pole two-loop integral in Section 6. These are narrow model-building and matching results within the established discrete-gauge axion literature [6, 7, 8, 9], not a derivation of the full ERT parameter point from one scale and not experimental evidence for an axion.

The continuous embedding passes its stated tests: all local anomalies vanish, renormalizable PQ remains accidental, the axion survives the gauge projection, and the explicitly evaluated unit-coefficient quality terms are far below the neutron-EDM requirement. Every anomalon pair now has a renormalizable ordinary-matter portal, and a generic mass matrix leaves three chiral families. The completion also changes the EFT interpretation: the old two-loop graph is $P = 16$, while the decay-enabled heavy matter opens a $P = 8$ two-loop closure. Its exact benchmark shift is nevertheless four orders below the direct dimension-21 scalar term, which dominates the calculated terms. Unknown Wilson/flavour/Yukawa tensors remain parameters, so "full amplitude" here means the complete finite momentum kernel and mass dependence for the stated graph, not a coefficient-free prediction.

The v16–v20 audit also records negative results: the v15 dimension-9 one-insertion potential and its $\mathbb{Z}_3$ cure were incorrect, and the alternate v18 scalar-only UV-quality proof is incomplete. V19's exact stable-anomalon caveat is resolved rather than hidden: its charge set is replaced. The remaining scientific tasks are an independent representation-theory and diagrammatic review, broken-phase Clebsch and flavour fits, RG evolution of the Wilson tensors, a radiatively stable v_Φ/v_S hierarchy, a thermal-restoration calculation, an ultimate quantum-gravity origin, and the nonstandard string-network calculation. The release retains the v17 65-check and v19 59-check regression engines and adds a v20 40-check engine with a controlled nonzero-exit path and machine-readable verdict.

A Reproduction formulas

The rephasing diagnostic referenced in Section 4 is

$$M = \begin{pmatrix} 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ -3 & 1 & 0 & 0 & 0 & 1 \\ 4 & 1 & 1 & 0 & 1 & 0 \\ -2 & 1 & 0 & 0 & 1 & 2 \\ -2 & 0 & 0 & 0 & 4 & 0 \end{pmatrix} \tag{29}$$

over $(X_S, X_F, X_{10}, X_{\overline{126}}, X_a, X_b)$, rows: the two Yukawas, spectator mass, ERT lock, $(S^\dagger)^3 16_F \overline{16}_s$, $S^4 16_F 16_s 10_H$, $(S^\dagger)^2(16_F\overline{16}_s)(16_s\overline{16}_s)$, $(S^\dagger)^2(16_s 16_s)_{10}(16_s 16_s)_{10}$. The proton width used in the Monte Carlo is

$$\Gamma(p \to e^+\pi^0) = \frac{m_p}{32\pi}\left[1 - \frac{m_\pi^2}{m_p^2}\right]^2 \left(\frac{4\pi\alpha_{\rm GUT}}{M_X^2}\right)^2 A_R^2\,|W_0|^2\left[1 + (1+|V_{ud}|^2)^2\right], \tag{30}$$

with W_0 from lattice QCD [21] and $\tau_p = \hbar/\Gamma$.

References

[1] A. Ernst, A. Ringwald and C. Tamarit, JHEP **02** (2018) 103, arXiv:1801.04906.

[2] C.-T. Hsieh, arXiv:1808.02881.

[3] L. M. Krauss and F. Wilczek, Phys. Rev. Lett. **62** (1989) 1221.

[4] T. Banks and M. Dine, Phys. Rev. D **45** (1992) 1424, arXiv:hep-th/9109045.

[5] L. E. Ibáñez, Nucl. Phys. B **398** (1993) 301, arXiv:hep-ph/9210211.

[6] K. S. Babu, I. Gogoladze and K. Wang, Phys. Lett. B **560** (2003) 214, arXiv:hep-ph/0212339.

[7] A. G. Dias, V. Pleitez and M. D. Tonasse, Phys. Rev. D **67** (2003) 095008, arXiv:hep-ph/0211107.

[8] L. Di Luzio, JHEP **11** (2020) 074, arXiv:2008.09119.

[9] K. S. Babu, B. Dutta and R. N. Mohapatra, Phys. Rev. Lett. **134** (2025) 111803, arXiv:2410.07323.

[10] K. S. Babu, B. Dutta and R. N. Mohapatra, "Accidental Peccei–Quinn Symmetry From Gauged U(1) and a High Quality Axion," arXiv:2412.21157.

[11] Q. Bonnefoy, Phys. Rev. D **108** (2023) 035023, arXiv:2212.00102.

[12] A. Djouadi, R. Fonseca, R. Ouyang and M. Raidal, Eur. Phys. J. C **83** (2023) 654, arXiv:2212.11315.

[13] D. Chang, T. Fukuyama, Y.-Y. Keum, T. Kikuchi and N. Okada, "Perturbative SO(10) Grand Unification," arXiv:hep-ph/0412011.

[14] Q. Lu, M. Reece and Z. Sun, JHEP **07** (2024) 227, arXiv:2312.07650.

[15] T. Kobayashi, R. Kurematsu and F. Takahashi, JCAP **09** (2013) 032, arXiv:1304.0922.

[16] P. W. Graham and D. Racco, JHEP **12** (2025) 028, arXiv:2506.03348.

[17] J. Benabou, M. Buschmann, J. W. Foster and B. R. Safdi, arXiv:2412.08699.

[18] K. Saikawa, J. Redondo, A. Vaquero and M. Kaltschmidt, "Spectrum of global string networks and the axion dark matter mass," JCAP **10** (2024) 043, arXiv:2401.17253.

[19] S. Borsányi et al., "Calculation of the axion mass based on high-temperature lattice quantum chromodynamics," Nature **539** (2016) 69, arXiv:1606.07494.

[20] G. Grilli di Cortona et al., JHEP **01** (2016) 034, arXiv:1511.02867.

[21] J.-S. Yoo et al., Phys. Rev. D **105** (2022) 074501, arXiv:2111.01608.

[22] DALI Collaboration, arXiv:2603.21951 (2026).

[23] Comment on the DALI sensitivity, arXiv:2607.28095 (2026).

[24] L. E. Ibáñez and G. G. Ross, "Discrete gauge symmetries and the origin of baryon and lepton number conservation," Nucl. Phys. B **368** (1992) 3 (mixed-gauge discrete anomalies).

[25] CAST Collaboration, Phys. Rev. Lett. **133** (2024) 221005, arXiv:2406.16840.

[26] P. Brun et al. (MADMAX), Eur. Phys. J. C **79** (2019) 186.

[27] A. J. Millar et al., Phys. Rev. D **107** (2023) 055013, arXiv:2210.00017.

[28] B. T. McAllister et al., Phys. Dark Univ. **18** (2017) 67, arXiv:1706.00209.

[29] A. Takenaka et al. (Super-Kamiokande), Phys. Rev. D **102** (2020) 112011.

[30] J. Bian et al. (Hyper-Kamiokande), arXiv:2203.02029.